QP CODE: 24900183

Reg No:.....

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course - MG1DSCMOS100 - PC HARDWARE AND SMARTPHONE TROUBLESHOOTING

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A

Multiple Choice Questions

Answer all questions.Each question carries 1 mark

1 Which part of the PC is responsible for temporarily storing data and programs that are actively used? [U] [1]

a) CPU b) RAM

c) Hard Drive d) Power Supply

2 What is the primary function of a socket on a motherboard? [U] [1]

a) Connects external peripherals b) Holds the CPU

c) Provides power d) Manages memory

3 What is the primary function of the CPU in a smartphone motherboard? [K] [2]

a) Manage battery life b) Process instructions

c) Store data d) Connect to Wi-Fi

4 If a flowchart-based troubleshooting process leads to no solution after several attempts, what should be the next step? [A] [2]

a) Escalate to a more advanced support team
b) Abandon the troubleshooting process
c) Keep repeating the same steps
d) Ignore the issue

5 If an SMPS fails to power on, what could be a simple diagnostic step? [U] [2]

a) Replacing the motherboard
b) Shorting the green and black wires
c) Checking the fan
d) Replacing the CPU

6 What might be the cause if a system fails to load the OS after a BIOS update? [A] [2]

a) Incorrect OS configuration
b) Corrupted OS files
c) The boot mode is set incorrectly
d) Insufficient RAM capacity

7 What is the primary purpose of using a flowchart in troubleshooting methodologies? [U] [2]

a) To document the problem
b) To provide a visual guide for decision-making
c) To track the time spent on troubleshooting
d) To replace technical manuals

8 Which component is responsible for rendering graphics in a smartphone? [K] [3]

a) CPU
b) RAM
c) GPU
d) RF SECTION

9 How does the PMIC contribute to battery life? [U] [3]

a) By providing high voltage output
b) By regulating power consumption for various components
c) By increasing RAM capacity
d) By enhancing graphical performance

10 In a mobile game that requires tilting the device to control movement, which sensors are primarily utilized? [A] [2]

a) Proximity sensor and magnetometer
b) Accelerometer and gyroscope
c) Light sensor and camera
d) Acetometer and temperature

sensor

(10 × 1 = 10)

Part B

Short Answer Questions

Answer 4 questions.Each question carries 5 marks

11	Draw the block diagram of a PC and give its working.	[K]	[1]
12	Explain how the voltage regulator, BIOS ROM, CMOS battery, and RAM work together to ensure the smooth operation of a computer.	[U]	[1]
13	What is the purpose of the RF unit on a smartphone motherboard?	[K]	[2]
14	Name three common types of connectors found on the back panel of a motherboard.	[U]	[2]
15	Compare the functions of the accelerometer and gyroscope in a smartphone.	[U]	[2]
16	Describe how GPS (Global Positioning System) functions in a smartphone.	[U]	[2]

(4 × 5 = 20)

Part C

Essay Questions

Answer 2 questions.Each question carries 10 marks

17	Explain the working principle of a Switched Mode Power Supply unit in a PC.	[U]	[1]
18	What is a chipset? Explain the working of each component of a chipset.	[U]	[2]
19	Explain the role of the RF Unit in smartphones and Discuss the types of antennas used in smartphones,	[A]	[2]
20	Draw and explain the typical functional diagram of smart phone	[U]	[3]

(2 × 10 = 20)

END OF THE QUESTION PAPER

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